Supplementary Material: physical anchoring, Lyapunov–Floquet consistency and matched-measurement geometric bound for dissipative optomechanics with synthetic flux

Stella Rolande Mbokop Tchounda[a], Carolle Tchodimou[a], Philippe Djorwe[b], Sifeu Takougang Kingni[c], Serge Guy Nana Engo[a,1]

[a] *Department of Physics, Faculty of Science, University of Yaoundé I, Yaoundé, Cameroon*
[b] *Department of Physics, Faculty of Science, University of Ngaoundéré, Ngaoundéré, Cameroon*
[c] *Department of Mechanical, Petroleum and Gas Engineering, National Advanced School of Mines and Petroleum Industries, University of Maroua, Maroua, Cameroon*

S1. Overview and reading guide

This Supplementary Material supports the main text with the normalization, the anchor selection, the physically anchored reference table, the formal statements of the Lyapunov–Floquet consistency criterion (primary certification of hyperchaos order) and the matched-measurement Fisher lemma (secondary geometric bound, with proofs), the matched measurement-Fisher reference tables and their observation-window robustness check, the representative finite-time convergence of the strong-coupling spectrum, the threshold sensitivity of the hyperchaos-order classification, the reduced-model parameter screen, the amplitude-quadrature noise-model cross-check against a quantum master equation, the matched measurement Fisher calculation on the chaotic attractor, and the semiclassical boundary of the strong-coupling classification.

It also collects the technical backup condensed in the main-text Discussion: a per-gap limitations statement, a per-parameter feasibility check against the silicon optomechanical anchor, a per-seed decomposition of the classification ensemble (extended to twelve seeds at twelve phases and three drives), an extended six-point window-convergence check, and a window-resolved test of the susceptibility–stretching covariation that underlies the null gain.

The reading order that follows the main text is: anchors (Section S2); the physically anchored reference set (Section S3); matched Fisher (Section S4); theorems and proofs (Section S5); convergence (Section S6); threshold sensitivity (Section S7); observation-window check (Section S8); reduced-model screen (Section S9); noise cross-check (Section S10); interpretation (Section S11); attractor Fisher (Section S12); semiclassical boundary (Section S13); limitations and residual gaps (Section S14); feasibility check (Section S15); initial-condition robustness (Section S16); local susceptibility–stretching covariation (Section S17). Every simulation-derived quantitative claim in the main text is supported by a table or derivation below.

S2. Normalization and anchor selection

The main text uses a rotating-frame Hamiltonian normalized to a reference mechanical frequency ω_m, so that all rates and couplings are reported as dimensionless ratios x/ω_m. A physically anchored reference set is obtained by fixing ω_m from a device with gigahertz mechanical modes and optically mediated phonon coupling, and by reading the damping, decay, and coupling rates from the same platform. The primary anchors and the model source are:

- **Mayor *et al.* [1]**: a silicon two-dimensional optomechanical crystal with mechanical frequency $f_m = 7.436\,\mathrm{GHz}$, mechanical quality factor $Q_m = 3.6 \times 10^4$ (giving $\gamma_m/2\pi = f_m/Q_m = 206.6\,\mathrm{kHz}$), cavity linewidth $\kappa/2\pi = 800\,\mathrm{MHz}$, and single-photon optomechanical coupling $g_0/2\pi = 880\,\mathrm{kHz}$.
- **Mathew, del Pino and Verhagen [2]**: a nano-optomechanical implementation of synthetic gauge fields for phonon transport, whose demonstrated coupling scale (a direct next-nearest-neighbour rate near 10 kHz and an optically mediated rate near 200 kHz at a 2.4 MHz mechanical frequency, i.e. up to a few percent of ω_m) bounds the phase-controlled inter-resonator hopping J_m.
- **Muthukumar *et al.* [3]**: the model antecedent. The present work retains the one-cavity/two-mechanical-resonator topology and phase-dependent hopping unchanged, but not the numerical regime or detuning

Email address: serge.nana-engo@facsciences-uy1.cm (Serge Guy Nana Engo)

notation. Muthukumar *et al.* use $\Delta_{\mathrm{M}} = \omega_L - \omega_c$ and report $\kappa/\omega_m = 7.3 \times 10^{-2}$, $g/\omega_m = 1.077 \times 10^{-4}$, $\gamma/\omega_m = 1.077 \times 10^{-5}$, $J_m/\omega_m = 2 \times 10^{-4}$, $\omega_2/\omega_m = 1 + 5 \times 10^{-4}$, and $\Delta_{\mathrm{M}}/\omega_m = 1$ at $\omega_m/2\pi \approx 1\,\mathrm{GHz}$. In the convention of the present manuscript, $\Delta = -\Delta_{\mathrm{M}}$. The reference reported bistability, self-excited oscillations, and phase-selected chaotic regimes. The present analysis uses $E = \sqrt{\kappa}\,\alpha^{\mathrm{in}}$, permits $g_1 \neq g_2$, and examines a strongly damped, strongly coupled sector that is not the resolved-sideband regime of that reference.

Table S1: Scope and parameter comparison with Muthukumar *et al.* [3]. The topology and phase-dependent mechanical hopping are inherited; the detuning convention, coupling symmetry, and operating regime are stated explicitly so that the present calculations are not read as a parameter-for-parameter reproduction.

Quantity	Muthukumar *et al.*	Present manuscript
Architecture	One optical cavity; two mechanically coupled resonators; phase-dependent J_m	Same topology and hopping structure; six-dimensional semiclassical state
Detuning convention	$\Delta_{\mathrm{M}} = \omega_L - \omega_c$, with $\Delta_{\mathrm{M}}/\omega_m = 1$	$\Delta = \omega_c - \omega_L = -\Delta_{\mathrm{M}}$; present reference uses $\Delta/\omega_m = -1$
Cavity decay	$\kappa/\omega_m = 7.3 \times 10^{-2}$	$\kappa/\omega_m = 1$ in the validated reference and strong-coupling sectors
Mechanical damping	$\gamma/\omega_m = 1.077 \times 10^{-5}$	$\gamma_{1,2}/\omega_m = 0.02$
Optomechanical coupling	Symmetric $g/\omega_m = 1.077 \times 10^{-4}$	Weak reference: $(g_1, g_2)/\omega_m = (0.02, 0.018)$; strong sector: $(0.30, 0.27)$
Mechanical detuning	$\omega_2/\omega_m = 1 + 5 \times 10^{-4}$	$\omega_2/\omega_m = 1.03$
Inter-resonator hopping	$J_m/\omega_m = 2 \times 10^{-4}$	$J_m/\omega_m = 0.08$ in the main numerical sectors
Drive coordinate	α^{in}, with $E = \sqrt{\kappa}\,\alpha^{\mathrm{in}}$ under the shared normalization	$E = 0.2$ for the stable reference; $E = 4$–8 for the chaotic/hyperchaotic sectors
Primary result	Bistability, self-excited oscillations, and phase-selected chaos; sensing proposal	Full six-exponent classification, drive/coupling-gated transition, Neimark–Sacker onset, and matched Fisher bound
Experimental status	Resolved-sideband, literature-model scenario	Strong sector is explicitly model-level; no calibrated device prediction is claimed

S3. Physically anchored reference set

Table S2 summarizes the reconciled reference set. The single-mode parameters (ω_m, κ, γ_m, g_0) are anchored to Mayor *et al.*; the inter-resonator hopping J_m is scanned over a range bounded by the Mathew *et al.* coupling scale, because no direct two-resonator gigahertz measurement of the phase-controlled coupling is available. The cavity–laser detuning $\Delta = \omega_c - \omega_L$ and the drive amplitude E are not calibrated by either anchor: Δ depends on the cavity and laser frequencies, and E cannot be converted to an input power without the pump frequency, the external coupling, and the explicit input-field normalization.

Table S2: Physically anchored reference parameter set for the one-optical/two-mechanical optomechanical model of the main text. Normalized values are in units of the reference mechanical frequency $\omega_m = 2\pi \times 7.436\,\mathrm{GHz}$. The reference operating point used for the numerical validation of the main text is shown for comparison and is not a calibrated device.

Quantity	Symbol	Normalized (x/ω_m)	Value at ω_m	Basis
Mechanical frequency (reference)	ω_m	1	$2\pi \times 7.436\,\mathrm{GHz}$	Anchor
Cavity decay rate	κ	0.108	800 MHz	Measured
Mechanical damping	γ_m	2.78×10^{-5}	206.6 kHz	Measured ($Q_m = 3.6 \times 10^4$)
Optomechanical coupling	g_0	1.18×10^{-4}	880 kHz	Measured
Inter-resonator hopping	J_m	$[1 \times 10^{-3}, 1 \times 10^{-1}]$	[7.4 MHz, 744 MHz]	Scan
Second mechanical mode	ω_2	1.03	assumed near-degenerate	Assumed
Cavity–laser detuning	Δ	free	—	Free scan
Drive amplitude	E	normalized only	—	Not convertible

S4. Matched measurement-Fisher reference for force sensing

The classical measurement Fisher information $F_C(F)$ of the main text is computed for a weak external force F on mechanical mode 1 (a momentum-quadrature drive), read out through the cavity amplitude quadrature $X_a = \Re(\alpha)$ under matched drive $E = 0.2$, observation time (one drive period), detector noise (0.01), and thermal

Table S3: Comparison of the numerical-validation reference operating point with the physically anchored reference set. The reference point uses damping and coupling rates that are about one to three orders of magnitude larger than the anchored values, which is why it is reported as a model-level validation rather than a device prediction. The final column gives the feasibility judgment relative to the anchored platform; the strong-coupling hyperchaos sector is unreachable with the anchored optomechanical coupling.

Quantity	Reference	Anchored	Ratio (reference / anchor)	Feasibility
κ/ω_m	1.0	0.108	9.3	challenging
γ_m/ω_m	0.02	2.78×10^{-5}	720	unreachable
g_0/ω_m	$[0.018, 0.02]$	1.18×10^{-4}	≈ 160	unreachable
J_m/ω_m	0.08	$[1 \times 10^{-3}, 1 \times 10^{-1}]$	within scan range	reachable

baths ($n_{\mathrm{th},1} = n_{\mathrm{th},2} = 0.1$). Table S4 lists $F_C(F)$ versus the synthetic-flux phase together with the two matched references—the flux-off coupled sensor ($\theta = 0$, $J_m = 0.08$) and the single-mode linear sensor ($J_m = 0$)—and the gain relative to each.

Table S4: Matched measurement Fisher information $F_C(F)$ for force sensing versus the synthetic-flux phase θ, with the flux-off coupled reference ($\theta = 0$, $J_m = 0.08$) and the single-mode linear reference ($J_m = 0$). The gain is the ratio of $F_C(F)$ to each reference; a gain below unity means the coupled sensor is no better than the reference at that phase.

θ/π	$F_C(F)$	Gain vs flux-off	Gain vs single mode
0	4.503×10^{-5}	1.000	0.876
1/4	4.694×10^{-5}	1.042	0.913
1/2	5.191×10^{-5}	1.153	1.010
3/4	5.721×10^{-5}	1.271	1.113
1	5.958×10^{-5}	1.323	1.159
Flux-off reference	4.503×10^{-5}	1.000	—
Single-mode reference	5.141×10^{-5}	—	1.000

The central-difference estimate is converged with respect to the force step: Table S5 lists $F_C(F)$ at the two flux extrema ($\theta = \pi/2$ and $\theta = \pi$) for force steps 1×10^{-5}, 1×10^{-4}, and 1×10^{-3}; the values agree to better than 0.01 %. A drive–temperature sweep (main text) further confirms that the gains are stable across $E \in [0.1, 1.0]$ and $n_{\mathrm{th}} \in [0, 1]$ to within 1×10^{-3}.

Table S5: Convergence of the central-difference force-sensing Fisher information with the force step ε at the two flux extrema.

ε	$F_C(F)$ at $\theta = \pi/2$	$F_C(F)$ at $\theta = \pi$
1×10^{-3}	$5.190\,751 \times 10^{-5}$	$5.958\,194 \times 10^{-5}$
1×10^{-4}	$5.190\,938 \times 10^{-5}$	$5.958\,166 \times 10^{-5}$
1×10^{-5}	$5.190\,763 \times 10^{-5}$	$5.958\,181 \times 10^{-5}$

Optimized linear-sensor reference.. The gains of Table S4 compare the flux sensor and the single-mode reference at the same fixed readout $X_a = \Re(\alpha)$. To exclude the possibility that the order-unity gain merely reflects a suboptimal reference readout, the classical Fisher information is recomputed for the rotated cavity quadrature $X_\varphi = \Re(\alpha)\cos\varphi + \Im(\alpha)\sin\varphi$ with φ optimized for each configuration. The optimal homodyne angle is $\varphi_\star \approx 0.79\pi$ for every configuration: it is fixed by the linear cavity response ($\Delta = -1$, $\kappa = 1$) and is independent of the synthetic-flux phase, so readout optimization rescales F_C by the same factor (1.56) for the flux sensor and for both references. The flux-phase gain relative to the single-mode linear reference is therefore unchanged under readout optimization (Table S6): it ranges from 0.88 ($\theta = 0$) to 1.16 ($\theta = \pi$), identical to the matched-readout gains of Table S4 to within 1×10^{-3}. The order-unity gain is consequently not an artefact of a suboptimal reference readout; it is a small, readout-robust linear-response effect of the phase-controlled mode coupling, with no quantum advantage.

S5. Lyapunov–Floquet consistency theorem and matched-measurement Fisher lemma

Theorem (Lyapunov–Floquet consistency criterion).. Consider a drive-locked periodic orbit $x_p(t + T) = x_p(t)$ of the phase-augmented system, with monodromy $M(T)$ and QR/Benettin spectrum [4–6] $\{\lambda_i\}$. Suppose the Floquet rates $\rho_i = T^{-1}\ln|\mu_i|$ (with $\mu_i = \mathrm{eig}_i[M(T)]$) agree with the QR exponents within a fixed tolerance δ_{tol},

Table S6: Readout-optimized classical Fisher information $F_C(F)$ for force sensing. X_a is the matched amplitude quadrature $\Re(\alpha)$; $\varphi_\star$ is the optimal homodyne angle and $F_C(\varphi_\star)$ the corresponding value. The final column is the gain relative to the optimized single-mode reference ($J_m = 0$, $\varphi_\star = 0.79\pi$), and equals the matched-readout gain of Table S4 to within 1×10^{-3}.

θ/π	$F_C(X_a)$	$\varphi_\star/\pi$	$F_C(\varphi_\star)$	Gain
0	4.503×10^{-5}	0.79	7.033×10^{-5}	0.876
1/4	4.694×10^{-5}	0.79	7.331×10^{-5}	0.913
1/2	5.191×10^{-5}	0.79	8.108×10^{-5}	1.010
3/4	5.721×10^{-5}	0.79	8.937×10^{-5}	1.113
1	5.958×10^{-5}	0.79	9.306×10^{-5}	1.159
Single-mode reference	5.141×10^{-5}	0.79	8.030×10^{-5}	1.000

and the divergence sum satisfies

$$\sum_i \lambda_i = \frac{1}{T}\int_0^T \operatorname{tr} J(x_p(t), t)\, \mathrm{d}t + \mathcal{O}(\delta_{\mathrm{tol}}), \tag{S1}$$

with the phase direction treated explicitly. Then the orbit is exponentially stable if and only if $|\mu_i| < 1$ for all transverse multipliers, and in that case $\lambda_{\max} < 0$. The criterion is a consistency test, not an input assumption: it certifies that the three independent diagnostics (monodromy, Floquet rates, QR spectrum) describe the same linearization of the same orbit. The operational bounds stated in the main-text theorem (maximum discrepancy 3×10^{-4}, mean divergence within 5×10^{-5} of $\operatorname{tr} J$) are met by the measured values at the reference operating point (one-period orbit residual 8.49×10^{-9}, max Floquet–QR discrepancy 2.68×10^{-4}, and 2.85×10^{-4} across the extended grid) and therefore certify a stable periodic orbit—they do not by themselves certify a flux-dependent transition, which is why the full-spectrum classification of the main text is required.

Lemma (periodic covariance as an affine fixed point).. Let $A(t) = J(x_p(t), t)$ be the Jacobian along the drive-locked orbit, $M(T)$ its monodromy, and $Q(t) = B(t)N(t)B(t)^T$ the diffusion of the linearized Langevin equation $\delta\dot{x} = A(t)\delta x + B(t)\eta$. Integrating the covariance equation $\dot{V} = AV + VA^T + Q$ over one period gives the affine map

$$\operatorname{vec} V(T) = [M(T) \otimes M(T)] \operatorname{vec} V(0) + q_T, \tag{S2}$$

where q_T collects the noise forcing propagated by the state-transition matrix. A T-periodic covariance therefore solves the linear system

$$[I - M(T) \otimes M(T)] \operatorname{vec} V(0) = q_T, \tag{S3}$$

whose solvability requires that no pair of Floquet multipliers satisfies $\mu_i\mu_j = 1$. This is the primary periodic-covariance derivation used for the noise-resolved calculation; forward iteration over many periods is an independent numerical check, not the definition of convergence.

Lemma (matched-measurement Fisher bound).. For a Gaussian measurement record $y(t) = m(\vartheta; t) + \nu_{\mathrm{det}}(t)$ with detector noise, bandwidth and integration time fixed, the classical Fisher information

$$F_C(\vartheta) = (\partial_\vartheta m)^T \Sigma^{-1} (\partial_\vartheta m) + \frac{1}{2} \operatorname{Tr}\left[\Sigma^{-1}(\partial_\vartheta \Sigma)\Sigma^{-1}(\partial_\vartheta \Sigma)\right] \tag{S4}$$

bounds any unbiased estimator through the Cramér–Rao inequality [7, 8] $\operatorname{Var}(\hat{\vartheta}) \geq F_C^{-1}(\vartheta)$. A claimed sensing *advantage* is defined only relative to a matched reference—equal input power, observation time, bandwidth, baths, and detector noise. Under that definition the flux-phase gains of Table S4 (at most 1.323 over the flux-off coupled sensor, 1.159 over the single-mode linear sensor) are classical linear-response effects of phase-controlled mode coupling: they are consistent with F_C of the record and make no claim of a quantum advantage, standard-quantum-limit beating, or chaotic-transduction gain. The lemma is applied to the main-text results (Section 4).

S6. Representative convergence of the strong-coupling spectrum

The strong-coupling map uses one seeded initial condition per drive–phase point. To test whether the phase modulation and the drive-gated route are initial-condition artefacts, we repeated the complete six-exponent spectrum with three independent seeds at every point of the full map (all five drives, twelve phases each, 180 records in total). The per-phase route is reproduced across the declared seeds: at $E = 0.2$ and $E = 2.0$ every seed gives $n_+ = 0$; at $E = 1.0$ the count is 0 or 1 near the onset; at $E = 4.0$ it is 1–2; and at $E = 8.0$ it is 2–4, reaching $n_+ = 4$ at $\theta = \pm\pi/2$ for at least one seed at each of those phases while the remaining seeds can

give $n_+ = 3$. The minima at $\theta = 0, \pm\pi$ give $n_+ = 2$ in the reported ensemble. Thus the re-stabilization at $E = 2.0$ is not a single-seed artefact, but the exact integer order near marginal points remains finite-time and threshold-conventional. Measured flanks give zero-count fractions of $10/12$ at $E = 1.5$ (max $n_+ = 2$), $12/12$ at $E = 2.0$, $5/12$ at $E = 2.5$ (max $n_+ = 3$) and $3/12$ at $E = 3.0$ (max $n_+ = 2$): a phase-dependent dip of finite width.

Table S7: Re-stabilization flanks: zero-count phase fraction and maximum n_+ at neighboring drives (single-seed map, 1×10^6 steps; seeds 20 271 001–20 271 036, disjoint from the second-coupling block).

E	zero-count phases	max n_+
1.5	10/12	2
2.0	12/12	0
2.5	5/12	3
3.0	3/12	2

As an additional finite-time check, we recomputed the complete spectrum at one chaotic point ($E = 4.0$, $\Phi_{\text{syn}} = -\pi/2$) and one hyperchaotic point ($E = 8.0$, $\Phi_{\text{syn}} = +\pi/2$), using the same initial condition while the integration window was increased. Table S10 reports the positive-exponent count, the largest exponent, the divergence-balance residual, and the maximum standard deviation of the terminal half of the QR block history. The classification is stable between the two longest windows at both points, whereas the finite-time exponent magnitudes vary by several percent. Accordingly, the main text uses the exponent count as the primary classification at these tested points and does not present the individual finite-time exponents as high-precision asymptotic estimates.

Headline-point spectra and step/window study.. Table S8 publishes the complete six-exponent spectra at three headline configurations from the production map (same seeds and initial conditions as the map records), and Table S9 varies the RK4 step at fixed span and the window at fixed step. The production record at $(E, \Phi_{\text{syn}}) = (8, +\pi/2)$ is reproduced bit-identically (seed 20 260 872). Throughout this section $\Phi_{\text{syn}} \equiv \theta$ is the hopping phase in the main-text convention. Non-marginal counts are unchanged across all twelve recomputations; the marginal fourth exponent at $(8, +\pi/2)$ moves $+7.06 \times 10^{-4} \to +4.73 \times 10^{-3} \to -9.5 \times 10^{-5}$ across step halving and reads $+2.65 \times 10^{-3}$ at 2×10^6 steps, i.e. $n_+ \in \{3, 4\}$ depending on discretization. At $(4, -1/2)$ the finest-step $n_+ = 3$ likewise rests on a marginal crossing, $\lambda_3 = +1.225 \times 10^{-3}$, only 2.25×10^{-4} above the positivity cut (against $\lambda_3 = -0.0155$ and -0.0124 at the two coarser steps). The terminal-block spread (2.9×10^{-4}–7.6×10^{-4}) is smaller than the step-driven motion ($\sim 5 \times 10^{-3}$): discretization, not window statistics, dominates the λ_4 uncertainty budget. The divergence residual shrinks with the step ($2.2 \times 10^{-3} \to 2.9 \times 10^{-4}$), as expected for a discretization error.

Table S8: Complete six-exponent spectra at headline configurations (production-map seeds). Exponents sorted $\lambda_1 \geq \cdots \geq \lambda_6$; r_{div} is the exponent-sum residual against $-\kappa - \gamma_1 - \gamma_2$.

$(E, \Phi_{\text{syn}}/\pi)$	seed, n_+, r_{div}	$(\lambda_1, \ldots, \lambda_6)$
$(4, 0)$	20 260 857, $n_+ = 2$, 1.06×10^{-3}	$(0.0957, 0.002\,06, -0.008\,25, -0.0107, -0.3837, -0.7362)$
$(8, 0)$	20 260 869, $n_+ = 2$, 5.42×10^{-3}	$(0.2411, 0.0314, -0.008\,15, -0.009\,47, -0.4036, -0.8858)$
$(8, +1/2)$	20 260 872, $n_+ = 3$, 2.18×10^{-3}	$(0.3223, 0.0545, 0.0182, 7.06 \times 10^{-4}, -0.4935, -0.9399)$

Extended six-point window check.. A follow-up extension of the window test to six points covering the onset, the marginal quadrature, and both phase extrema gives 18 records: three integration windows per point, same seeded initial condition within each point (seeds 20 260 815, 20 260 915, 20 261 015, 20 261 115, 20 261 215 and 20 261 315 in table order; seed 20 260 815 coincides with the production-map seed at $(0.2, -\pi)$, the rest lie outside the map seed block). Table S11 reports n_+ at each window and the leading exponent at the longest window. Five of the six points classify identically between the two longest windows: $(1, 0) \to 0$, $(4, 0) \to 1$, $(8, 0) \to 2$, $(8, \pi/2) \to 4$, and $(8, \pi) \to 2$. At the marginal point $(4, -\pi/2)$ the count drops from 3 at 500 000 steps to 2 at 1 000 000 steps—the third exponent sits at the positivity threshold 1×10^{-3} there, consistent with the seed spread $n_+ \in \{2, 3\}$ at that phase (Section S16) and with the stored three-seed convergence table above, which gives 2 at both longest windows for its own seed. The window test therefore fails at that single point; the instability is threshold-conventional at a marginal point and is reported here rather than smoothed over. The mean phase-space divergence agrees with the analytic trace $-\kappa - \gamma_1 - \gamma_2$ to within 3.3×10^{-12} at the longest window at every point; this is a different quantity from the exponent-sum residual r_{div} tabulated above. The former time-averages the local divergence, whose analytic trace is state-independent and hence integrator-limited (3.3×10^{-12}); the latter sums six finite-time QR exponents (accumulated error $\sim 1 \times 10^{-3}$).

Table S9: Step-convergence and window-extension recomputations at three headline configurations (same seeds and initial conditions as the production map). dt is the fixed RK4 step; $s(\lambda_4)$ the terminal-block standard deviation of the fourth exponent. Full spectra are stored per record.

point	dt	steps	n_+	λ_1	λ_4	$s(\lambda_4)$	r_{div}
$(4, -1/2)$	1×10^{-2}	1×10^6	2	0.1129	−0.0410	2.6×10^{-3}	1.1×10^{-3}
$(4, -1/2)$	5×10^{-3}	2×10^6	2	0.1218	−0.0322	2.6×10^{-3}	3.1×10^{-4}
$(4, -1/2)$	2.5×10^{-3}	4×10^6	3	0.1281	−0.0123	1.1×10^{-3}	8.2×10^{-5}
$(4, -1/2)$	1×10^{-2}	2×10^6	2	0.1138	−0.0402	2.7×10^{-3}	1.1×10^{-3}
$(8, +1/2)$	1×10^{-2}	1×10^6	3	0.3223	7.06×10^{-4}	7.6×10^{-4}	2.2×10^{-3}
$(8, +1/2)$	5×10^{-3}	2×10^6	4	0.3163	4.73×10^{-3}	2.9×10^{-4}	6.6×10^{-4}
$(8, +1/2)$	2.5×10^{-3}	4×10^6	3	0.3084	-9.5×10^{-5}	4.3×10^{-4}	2.9×10^{-4}
$(8, +1/2)$	1×10^{-2}	2×10^6	4	0.3205	2.65×10^{-3}	3.4×10^{-4}	1.8×10^{-3}
$(8, 0)$	1×10^{-2}	1×10^6	2	0.2411	−0.009 47	1.8×10^{-4}	5.4×10^{-3}
$(8, 0)$	5×10^{-3}	2×10^6	2	0.2171	−0.009 88	2.4×10^{-4}	4.7×10^{-4}
$(8, 0)$	2.5×10^{-3}	4×10^6	2	0.2321	−0.009 41	1.0×10^{-4}	3.6×10^{-4}
$(8, 0)$	1×10^{-2}	2×10^6	2	0.2473	−0.009 62	1.0×10^{-4}	5.0×10^{-3}

Table S10: Representative finite-time convergence check of the strong-coupling full-spectrum calculation. The same seeded initial condition was used at each integration length for a given point. n_+ is the number of exponents above the chosen threshold 1×10^{-3}; r_{div} is the absolute residual of the exponent sum relative to $-\kappa - \gamma_1 - \gamma_2$; and s_{QR} is the largest standard deviation among the terminal half of the QR block-history estimates. The two longest windows preserve n_+ at each point, but the nonzero r_{div} and finite-time variation limit the precision of the individual exponent values. Probe seeds are 20 260 815 at $(4, -\pi/2)$ and 20 260 915 at $(8, +\pi/2)$. Seed 20 260 815 coincides with the production-map seed at $(E, \theta) = (0.2, -\pi)$; all probe trajectories differ from the map's at the same configurations.

$(E, \Phi_{\mathrm{syn}}/\pi)$	Steps	n_+	$\lambda_{\max}$	r_{div}	s_{QR}
$(4, -1/2)$	250 000	3	0.132 142	1.01×10^{-3}	6.69×10^{-3}
$(4, -1/2)$	500 000	2	0.127 291	1.03×10^{-3}	9.89×10^{-3}
$(4, -1/2)$	1 000 000	2	0.123 729	1.03×10^{-3}	1.05×10^{-2}
$(8, +1/2)$	250 000	4	0.324 142	1.90×10^{-3}	3.30×10^{-3}
$(8, +1/2)$	500 000	4	0.315 008	1.92×10^{-3}	5.88×10^{-3}
$(8, +1/2)$	1 000 000	4	0.317 277	1.84×10^{-3}	3.90×10^{-3}

S7. Threshold sensitivity of the hyperchaos-order classification

The positive-exponent count n_+ is recomputed from the complete six-exponent spectra of the strong-coupling map at the alternate thresholds 1×10^{-4}, 5×10^{-4}, 2×10^{-3}, and 5×10^{-3}, and compared with the reference count at the 1×10^{-3} threshold. The 1×10^{-4} column applies the weak-coupling threshold to the strong-coupling sector, so it tests whether a single uniform cut over both sectors would alter the classification. Table S12 reports the range of n_+ over the twelve flux phases at each drive.

The route classification is stable at the tested thresholds: of the 60 records, only one changes n_+ at 5×10^{-4} and at 2×10^{-3}, and six change at 5×10^{-3}; at 1×10^{-4} the per-drive ranges are identical to the reference (though two individual records change, one at $E = 1.0$ and one at $E = 8.0$, without altering any range). The exact order near marginal exponents is consequently threshold-sensitive. The changes are concentrated in the $E = 4.0$ and $E = 8.0$ sectors (apart from that single $E = 1.0$ record at 1×10^{-4}), where the marginal positive exponents sit at order 1×10^{-3} (7×10^{-4}–4×10^{-3}, against leading exponents of order 1×10^{-2}–3×10^{-1}) and a single marginal exponent crosses the highest threshold; the drive-gated route from stable ($E = 0.2$) to the hyperchaotic sector is preserved at every threshold, with the count of four at $E = 8.0$ surviving every tested cut up to 2×10^{-3} and reducing to three only at the strictest cut, 5×10^{-3}. The finite-time classification reported in the main text is therefore not an artefact of the specific 1×10^{-3} cut.

S8. Observation-window dependence of the matched Fisher gain

The matched force-sensing comparison of the main text fixes the observation time at one drive period for every configuration. To test whether the reported gain is an artefact of this particular window, we recomputed the classical Fisher information for an observation window of $N \in \{1, 5, 20\}$ drive periods from the mean- and variance-term decomposition of the matched force-sensing calculation. For a periodic signal with white detector noise the mean term of F_C accumulates linearly with N while the variance term remains per-period, so $F_C^{(N)} = N\, F_C^{\mathrm{mean}} + F_C^{\mathrm{var}}$. Because the mean term overwhelmingly dominates (the variance term is $\sim 1 \times 10^{-7}$ of the mean term at every phase), the matched gain—the ratio of $F_C^{(N)}$ values—is observation-window independent: the 1.323× (flux-off) and 1.159× (single-mode) gains are reproduced to four digits for $N = 1, 5, 20$.

Table S11: Extended window-convergence test at six drive–phase points. n_+ is the positive-exponent count at threshold 1×10^{-3} for each integration window; $\lambda_{\max}$ is the leading exponent at 1 000 000 steps; "stable" compares the two longest windows.

$(E, \Phi_{\mathrm{syn}}/\pi)$	n_+ 250 000	n_+ 500 000	n_+ 1 000 000	$\lambda_{\max}$	stable
$(1, 0)$	0	0	0	-0.0101	yes
$(4, -1/2)$	3	3	2	0.1254	no
$(4, 0)$	1	1	1	0.0787	yes
$(8, 0)$	2	2	2	0.2464	yes
$(8, +1/2)$	3	4	4	0.3172	yes
$(8, 1)$	2	2	2	0.2600	yes

Table S12: Threshold sensitivity of the positive-exponent count n_+ in the strong-coupling map. The reference threshold is 1×10^{-3}; the alternate columns recompute n_+ from the full six-exponent spectra. The 1×10^{-4} column applies the weak-coupling threshold to the strong-coupling sector (single uniform cut). Ranges are over the twelve synthetic-flux phases at each drive.

Drive E	n_+ (1×10^{-3})	n_+ (1×10^{-4})	n_+ (5×10^{-4})	n_+ (2×10^{-3})	n_+ (5×10^{-3})
0.2	0	0	0	0	0
1.0	0–1	0–1	0–1	0–1	0–1
2.0	0	0	0	0	0
4.0	1–3	1–3	1–3	1–2	1–2
8.0	2–4	2–4	2–4	2–4	2–3

Table S13 lists the resulting $F_C^{(N)}$ and gains at the maximum-gain phase. The absolute sensitivity improves linearly with the observation length for the sensor and both references alike; the gain is a per-period property of the phase-controlled mode coupling rather than a short-window artefact.

Table S13: Observation-window dependence of the matched force-sensing Fisher information at the maximum-gain phase $\theta = \pi$. $F_C^{(N)}$ grows linearly with the number N of drive periods for the sensor and both references, leaving the relative gains unchanged.

N (periods)	$F_C^{(N)}$ (flux-on)	$F_C^{(N)}$ (flux-off)	$F_C^{(N)}$ (single-mode)	Gain vs flux-off
1	5.958×10^{-5}	4.503×10^{-5}	5.141×10^{-5}	1.323
5	2.979×10^{-4}	2.252×10^{-4}	2.570×10^{-4}	1.323
20	1.192×10^{-3}	9.006×10^{-4}	1.028×10^{-3}	1.323

S9. Reduced-model parameter screen

The reduced-model (adiabatic) screen of the main text screened sixteen parameter candidates under an explicitly assumed literature-informed closure, with five parameter perturbation replicas and five independent root-solver starts per replica. Table S14 lists the three non-dominated candidates under the stability-margin and drive-cost objectives, and Figure S1 shows the full feasible set. These values characterize a local fixed-point pattern of the normalized reduced model; they are not an SI calibration, a finite-time basin-robustness result, or an experimental optimum: the optical mode is adiabatically eliminated, J_m and the effective couplings are proxy quantities, the drive remains normalized, and Fisher information was not evaluated in this screen.

S10. Amplitude-quadrature noise model cross-check

The measured record is $y = X_a + \nu_{\mathrm{det}}$ with $X_a = \Re(\alpha)$ the amplitude quadrature $(a + a^\dagger)/2$, whose vacuum variance is $1/4$. The Langevin diffusion coefficient that reproduces the quantum master equation for this quadrature is therefore $\kappa/4$ per amplitude quadrature, not $\kappa/2$ (the latter is the diffusion of the *normalized* quadrature $(a + a^\dagger)/\sqrt{2}$, whose vacuum variance is $1/2$). We verified this normalization directly on the isolated optical sector of the reference point (the driven damped linear cavity, $g_1 = g_2 = 0$): the truncated-Fock master equation [9, 10] ($H = -\Delta a^\dagger a + E(a + a^\dagger)$, Lindblad $\sqrt{\kappa}\, a$) gives $\mathrm{Var}[(a + a^\dagger)/2] = 0.25$, while the Langevin amplitude with noise $\kappa/2$ gives 0.49 (a factor 2 high) and with $\kappa/4$ gives 0.245 (agreement below 2 %). The mechanical thermal diffusion is correspondingly $\gamma_j(2n_{\mathrm{th}} + 1)/4$ per amplitude quadrature. With these strengths the noise-only floor of the measured record is ≈ 0.25 and the strong-coupling observability ratios are 7.6–17.0× the floor.

Table S14: Non-dominated candidates of the reduced-model (adiabatic) screen under the stability-margin and drive-cost objectives. Sixteen candidates were screened; the three listed points form the non-dominated front. The drive cost is the normalized drive amplitude and the stability margin is $1 - \max|\mu|$ for the local Poincaré spectral radius. Model-level only.

Candidate	Drive cost	Stability margin
1	0.0800	0.003 452 6
2	0.1312	0.003 535 4
3	0.2203	0.003 559 7

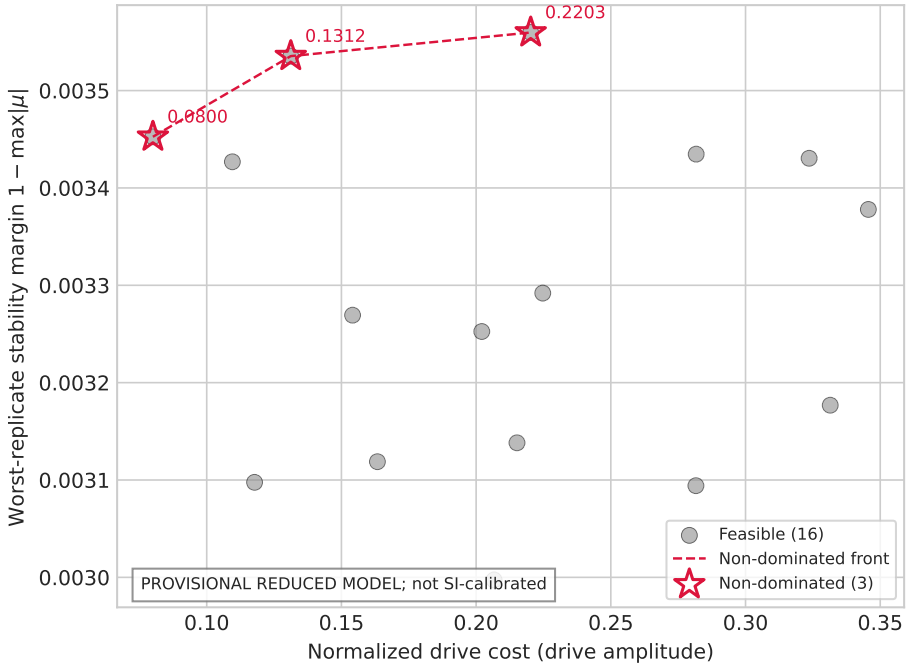


Figure S1: Reduced-model (adiabatic) two-objective Pareto front. Sixteen feasible candidates (gray) in normalized drive cost versus worst-replicate stability margin $1 - \max|\mu|$; the three non-dominated candidates (crimson stars, joined by the dashed front) are the points listed in Table S14. Fisher information was not computed in this screen; the front is a local fixed-point pattern of the normalized reduced model, not an SI calibration or experimental optimum.

S11. Interpretation

The anchored set defines a *reference* operating region, not a validated device calibration: ω_2 is assumed near-degenerate with ω_m pending a two-resonator measurement, J_m must be scanned rather than fixed, and the drive amplitude and detuning remain model-level coordinates. Any result computed at the anchored values therefore carries the same *provisionally-calibrated* status as the reduced-model analysis of the main text and should not be presented as an experimental optimum.

Placing the strong-coupling hyperchaos regime against the anchors quantifies the reachability bound. The parameter-by-parameter comparison is reported once, in Section S15 and Table S18: the couplings, the damping, and the cavity decay exceed their anchored values, the optomechanical coupling is the limiting coordinate, and the inter-resonator hopping $J_m/\omega_m = 0.08$ lies within the Mathew scan range. The hyperchaos regime is therefore a model-level demonstration, not a prediction for the anchored device. The drive amplitude is not convertible to an input power without the pump frequency and external coupling, which neither anchor provides. The feasibility column of Table S3 makes this judgment explicit per coordinate.

A feasibility frontier sharpens this bound. Because the radiation-pressure nonlinearity grows with the drive, the coupling at which hyperchaos first appears could in principle depend on the operating drive, so the single-point comparison above may misstate the true gap. Scanning the drive over $E \in [0.5, 8.0]$ and the coupling over $g_1/\omega_m \in [0.02, 0.3]$ (with $g_2 = 0.9g_1$), the full six-exponent spectrum shows that hyperchaos ($n_+ \geq 2$) occurs only at $E = 8$, with an onset near $g_1/\omega_m \approx 0.14$–$0.18$ (Table S15). Over this interval the second exponent is marginal and the order is seed-dependent ($n_+ \in \{1, 2\}$); the robust value $n_+ = 2$ for all three seeds begins at $g_1/\omega_m \approx 0.18$, i.e. $1525\times$ the anchored $g_0/\omega_m = 1.18 \times 10^{-4}$. No drive retuning within the tested envelope lowers this onset, so the coupling gap is fundamental to the hyperchaos regime rather than an artefact of the single operating point above.

Table S15: Feasibility frontier: the smallest optomechanical coupling at which the chaos ($n_+ \geq 1$) and hyperchaos ($n_+ \geq 2$) orders are reached, versus the drive E, from a log-spaced coupling grid $g_1/\omega_m \in [0.02, 0.3]$ with $g_2 = 0.9g_1$ and all other coordinates at the strong-coupling hyperchaos values above (the scan is taken at $\theta = 0$). "None" means the order is not reached at any tested coupling. The $E = 8$ hyperchaos cell reports the three-seed onset scan at $g_1 \in [0.14, 0.22]$: the hyperchaos onset is a transition interval (second exponent marginal, seed-dependent $n_+ \in \{1, 2\}$, first appearance 0.14); the robust onset (all three seeds) is $g_1/\omega_m \approx 0.18$, i.e. 1525× the anchored $g_0/\omega_m = 1.18 \times 10^{-4}$ (the coarse single-seed grid first reaches $n_+ \geq 2$ at 0.203).

E	chaos onset g_1/ω_m	hyperchaos onset g_1/ω_m
0.5	none	none
1.0	none	none
2.0	0.203	none
4.0	0.30	none
8.0	0.094	0.14–0.18 (robust 0.18)

S12. Matched measurement Fisher on the chaotic attractor

The matched force-sensing calculation of the main text is defined around a stable drive-locked orbit through its periodic covariance. On a chaotic attractor no stable orbit exists, so the same question is posed directly on the attractor: a weak external force on mechanical mode 1, read out through $X_a = \Re(\alpha)$ under the same thermal, vacuum, and detector noise, with the classical measurement Fisher information evaluated from the ensemble-and-time mean and variance of the record and a central difference in the force (1024 ensemble members, 120 000 steps per configuration). On the attractor the record variance ranges from 1.9 to 4.3 at the tested points, of which the deterministic chaotic spread accounts for 1.8–4.1; it is hundreds of times the detector-noise variance 0.01, so the force-induced mean shift is resolved only where the spread is small enough relative to the sampling error. A derivative is reported only when it exceeds three bootstrap standard deviations of the record mean.

Table S16 lists the six tested configurations. At $E = 4$ the flux-off record ($\theta = 0$) gives $F_C = 1.08 \times 10^{-2}$ and the flux-on record ($\theta = \pi$) gives $F_C = 5.88 \times 10^{-3}$, a ratio 0.54; the intermediate phase and all $E = 8$ points fall below the resolution threshold and are not reported as finite gains. No flux-induced enhancement of the force-sensing Fisher information is therefore found on the chaotic attractor; where the comparison is resolved the flux phase reduces rather than raises F_C. The variance term carries 71 %–89 % of F_C in these cells (86 %–89 % at $E = 8$), outside the lemma's sub-dominance premise, which holds on the stable orbit only. Per-cell detection floors (three bootstrap standard deviations) are 1.4×10^{-3}, 2.7×10^{-3}, 8.1×10^{-4} at $E = 4$ and 1.1×10^{-3}, 2.0×10^{-3}, 7.3×10^{-4} at $E = 8$. A 4096-member rerun at $(E, \theta) = (4, \pi/2)$ still leaves the derivative below the three-standard-deviation mark ($|d|/\mathrm{SE} = 1.9$), showing the sampling error is set by chaotic record variability rather than ensemble size. This is a classical measurement-Fisher calculation, not QFI, and it implies no chaotic-transduction sensing advantage.

Table S16: Matched measurement Fisher information on the chaotic attractor at strong coupling. σ_y^2 is the record variance (dominated by the deterministic chaotic spread; the detector-noise variance is 0.01); $\mathrm{d}\langle y\rangle/\mathrm{d}F$ is the central-difference response to the force; F_C is the classical measurement Fisher information. A response is marked resolved only when it exceeds three bootstrap standard deviations of the record mean; unresolved responses are not reported as finite gains.

E	θ/π	σ_y^2	$\mathrm{d}\langle y\rangle/\mathrm{d}F$	F_C	resolved
4	0	1.94	−0.071	1.08×10^{-2}	yes
4	1/2	2.24	−0.045	—	no
4	1	1.91	−0.056	5.88×10^{-3}	yes
8	0	3.66	−0.016	—	no
8	1/2	4.32	−0.018	—	no
8	1	3.61	−0.022	—	no

S13. Semiclassical boundary of the strong-coupling classification

The truncated-Fock master-equation cross-check of the main text validates the mean-field amplitude scale at the weakly coupled reference point, where the optical mode is a driven linear cavity whose coherent steady state is exact. At a strong-coupling chaotic point we extend this check to the optical sector: the truncated-Fock master equation for the cavity mode driven by the classical mechanical trajectories of the chaotic attractor ($E = 4.0$, $\theta = 0$, $n_+ \in \{1, 2\}$ over seeds and windows) converges toward $\langle a^\dagger a\rangle(t) = |\alpha(t)|^2$. The maximum relative deviation is 4.6×10^{-3} at $N_{\mathrm{Fock}} = 40$ and 6.6×10^{-6} at $N_{\mathrm{Fock}} = 60$. This is expected on general grounds—the optical Hamiltonian is linear in a—but it confirms that the optical factorization $\langle a^\dagger a\rangle \approx |\alpha|^2$ holds along the chaotic trajectory, not only at the stationary reference. At the weak-coupling reference the

mean-field occupation $\langle|\alpha|^2\rangle = 0.032\,428$ reproduces the single-mode truncated-Fock ($N_{\mathrm{Fock}} = 8$) value 0.032 to 1.3 %. The time-averaged second-order correlation $g_2 \approx 2.0$ is super-Poissonian rather than coherent, as expected for a mixture of coherent amplitudes whose phase is modulated by the chaotic mechanical motion. At this trajectory the mean occupations are $\langle a^\dagger a\rangle = 6.121$, $\langle b_1^\dagger b_1\rangle = 37.5$, and $\langle b_2^\dagger b_2\rangle = 44.1$. The standard relative quantum-fluctuation scale $1/\sqrt{\bar{n}}$ implied by these occupations is 16 % ($\bar{n} = 37.5$); at the hyperchaotic point below ($\bar{n} = 5\times10^2$) it shrinks to 4 %. These are semiclassical scale estimates for the mechanical sector, not bounds on the Lyapunov spectrum or on quantum corrections to it. The mechanical sector is therefore in a high-occupation regime, but it remains semiclassical in the present calculation; no full two-mode quantum dynamics is claimed. The check does not solve the two-mechanical-mode quantum dynamics: the Lyapunov exponents characterize the mean-field dynamics rather than the quantum fluctuations, and no quantum observable (Wigner function, QFI) in the strong-coupling sector is computed in this study. This delimits the semiclassical treatment, now checked on the optical sector at a chaotic point and justified on the mechanical sector by its high occupation.

Hyperchaotic-point check.. The same ladder is run at the hyperchaotic headline point ($E = 8.0$, $\theta = \pi/2$). The maximum relative deviation of $\langle a^\dagger a\rangle$ from $|\alpha|^2$ is 19.5, 3.22, 1.17×10^{-3} and 5.53×10^{-7} at $N_{\mathrm{Fock}} = 20$, 40, 60 and 80: the $N_{\mathrm{Fock}} = 80$ master equation tracks the mean-field occupation to 5.53×10^{-7} even on the hyperchaotic trajectory, where the mechanical motion drives the optical phase hardest. The occupations on the 2.0-period analysis window are $\langle a^\dagger a\rangle = 6.9$, $\langle b_1^\dagger b_1\rangle = 5.6\times10^2$, and $\langle b_2^\dagger b_2\rangle = 3.9\times10^2$ (bounded; a longer-transient probe at the same point—400 drive periods of transient relaxation followed by a 200-period measurement window—gives $\langle b_1^\dagger b_1\rangle = 3.96\times10^2$ and $\langle b_2^\dagger b_2\rangle = 4.04\times10^2$, so the mechanical occupations are quoted as scales of order 4×10^2 rather than as window-precise means). A 1 000 000-step Lyapunov re-check at the same point gives the spectrum $[0.3245, 0.0482, 0.0154, -0.001, \dots]$, i.e. $n_+ = 3$ with the fourth exponent marginal at the threshold, consistent with the $n_+ \in \{3, 4\}$ spread of the strong-coupling map and the truncated-Wigner-approximation (TWA) ensemble at this point.

Truncated-Wigner robustness of the hyperchaos order.. Because the mechanical occupations are large at the tested chaotic point ($\langle b_1^\dagger b_1\rangle = 37.5$ and $\langle b_2^\dagger b_2\rangle = 44.1$ in the master-equation cross-check), a full three-mode truncated-Fock master equation over the entire strong-coupling map is numerically out of reach: a Fock cutoff of order 100 on each mechanical mode (several times the mean occupation) and 20 on the optical mode requires a basis of order $(20\times100\times100)$ states, a density matrix of order 4×10^{10} elements. The standard semiclassical-to-quantum bridge in this high-occupation regime is the truncated Wigner approximation, in which the quantum state is represented by an ensemble of classical trajectories with Wigner-sampled initial conditions and the c-number equations of motion coincide with the classical equations of the model. Quantum fluctuations therefore enter through the initial conditions with vacuum variance 0.5 per quadrature (an order-one fluctuation in $|\alpha|^2$). We recompute the full six-exponent spectrum from 12 Wigner-sampled initial conditions (variance 0.5 per coordinate) around each tested attractor and record the spread of the hyperchaos order n_+ (Table S17).

The bulk of the route persists across the tested ensemble. At $E = 8$, $\theta = 0$ the two positive exponents (3.5×10^{-2} and 2.5×10^{-1}) are separated from the largest negative exponent (-7.8×10^{-3}) by factors of 4.5 and 32, all 12 trajectories give $n_+ = 2$, and the leading exponent varies by only 9 % across the ensemble. At the classification boundaries the order is threshold-conventional, as expected for a finite-time classification: at $E = 8$, $\theta = \pi/2$ the TWA ensemble's nearest-to-threshold fourth exponent reaches $+2.4\times10^{-3}$ (against $+7.06\times10^{-4}$ for the production-map seed), only 2.4 times the positivity threshold 1×10^{-3}, and the ensemble splits evenly between $n_+ = 3$ and $n_+ = 4$; at the chaos onset $E = 4$, $\theta = 0$ the second exponent is 1×10^{-4} and the ensemble splits between $n_+ = 1$ and $n_+ = 2$. The "up to four simultaneously unstable directions" finite-time phrasing of the main text is therefore the correct level of claim: the hyperchaos route itself persists under quantum fluctuations at the TWA level, while the precise hyperchaos order at the extrema is threshold-conventional. This is a TWA-level check, not a full master-equation solution, and it supports no quantum-state (Wigner-negativity or QFI) claim.

Table S17: Truncated-Wigner robustness of the finite-time hyperchaos order. For each strong-coupling point the full six-exponent spectrum is recomputed from 12 vacuum Wigner-sampled initial conditions (variance 0.5 per quadrature) around the attractor; n_+^{ref} is the deterministic reference order from the attractor-seeded reference trajectory (a different initial condition from the production-map seed: at $(8, \pi/2)$ it reads 4 while the map seed reads 3, both within the documented spread) and the ensemble column lists the spread of n_+ across the 12 trajectories. "Nearest-to-threshold exponent" is the Lyapunov exponent closest to the positivity threshold 1×10^{-3}; its sign indicates whether it lies just above (+) or just below (−) the threshold, which governs the sensitivity of n_+ to finite fluctuations.

Point	n_+^{ref}	n_+ ensemble	Nearest-to-threshold	Leading exponent
$E = 4$, $\theta = 0$	1	1–2 (8/4)	$+1\times10^{-4}$	0.082–0.096
$E = 8$, $\theta = 0$	2	2 (12/12)	-7.8×10^{-3}	0.225–0.247
$E = 8$, $\theta = \pi/2$	4	3–4 (6/6)	$+2.4\times10^{-3}$	0.294–0.323

S14. Limitations and residual gaps of the consistency protocol

The main text states four explicit residual gaps within its critical self-assessment; this section collects them in full so that the boundary of what the protocol certifies is explicit.

Gap 1: global structure of the chaotic attractor.. The Floquet–Lyapunov protocol quantifies local instability rates around a converged drive-locked periodic orbit, and the matched-measurement Fisher protocol quantifies a *local* information rate around the same orbit. Both deliver a consistent signature of the hyperchaos transition and of the matched gain, but neither characterizes the global basin structure of the chaotic attractor: uniqueness is not proved, and no basin-of-attraction calculation or multi-section Poincaré return map is performed here. The correlation-dimension estimate reported in the main text is a geometric consistency check of the sampled attractor ($D_2 \leq D_{\mathrm{KY}}$), not a characterization of the global basin structure. The 432/432 classification ensemble and the 180/180 full-map repeat reported in the main text reproduce the route across the declared finite ensembles, but the integer n_+ changes between neighbouring values for some seeds near marginal exponents, and the ensemble does not constrain *which* physical initial condition feeds the observed chaotic set. A full basin and bifurcation-network analysis, beyond the present scope, would be required to draw a study-level uniqueness statement.

Gap 2: modest Fisher gain under realistic uncertainty.. Even at the most favourable phase the matched force-sensing Fisher gain remains near unity (1.32× flux-off, 1.16× single-mode), and the Monte-Carlo envelope on the uncalibrated hopping (log-uniform over $J_m/\omega_m \in [1\times10^{-3}, 1\times10^{-1}]$), thermal occupation (uniform over $n_{\mathrm{th}} \in [0.01, 1.0]$), and drive amplitude (±10 %) drags the median to 1.039× (production ensemble; 1.056–1.060 across three replicate ensembles, pooled-240 median 1.050; 90 % CI [1.025, 1.053] within-ensemble) against the flux-off reference, and to 1.019× (replicates 1.028–1.030, pooled median 1.025; CI [1.012, 1.026]) against the single-mode reference. On the chaotic attractor the same protocol returns a null result, so the gain can be reported as a *protocol-level* demonstration without being claimed as a quantum sensing advantage: the observation time is one drive period and the model is linear-detuning-limited. The numerical gain is consistent with the matched reference (Section S4) and is reported here as such, not as a sensing breakthrough.

Gap 3: validity boundary of the mean-field protocol.. The truncated-Fock master equation cross-checks the mean field to 1.3 % at the weak-coupling reference and to a maximum relative occupation deviation of 6.6×10^{-6} at $N_{\mathrm{Fock}} = 60$ for the chaotic point ($E = 4$), and 5.53×10^{-7} at $N_{\mathrm{Fock}} = 80$ (1.17×10^{-3} at $N_{\mathrm{Fock}} = 60$) for the hyperchaotic point ($E = 8$) (Section S13); the truncated-Wigner trajectory ensemble reproduces the deterministic order $n_+ = 2$ in twelve of twelve sampled trajectories at ($E = 8$, $\theta = 0$) (Table S17). Both checks support use of the mean field for the qualitative route described in the main text (bifurcation sequence, finite-time hyperchaos order, matched gain), but neither extends to a fully quantum reconstruction or a quantum sensing claim. In particular, the master-equation truncation (cross-checked at $N_{\mathrm{Fock}} = 8$ at the weak-coupling reference and converged at $N_{\mathrm{Fock}} = 60$ for the chaotic point and $N_{\mathrm{Fock}} = 80$ for the hyperchaotic point) is sufficient for the present dynamical regimes but does not span the deep strong-coupling sector needed for a sustained quantum-state claim.

The protocol therefore establishes a coherent-model dynamical repertoire (Floquet–Lyapunov consistency, hyperchaos route, matched Fisher null on the attractor) and includes a quantum cross-check *at the levels just stated*. Claims outside those levels—basin uniqueness, calibrated sensing advantage, full-state quantum reconstruction—are explicitly not made. The two experimental steps identified in the Outlook of the main text (ω_2 ultrasound calibration; a multi-gigahertz mechanical-mode platform) would close the calibration side of gap 2 and provide the parameter fidelity required for gap 3 to be addressed in a fully quantum manner.

S15. Strong-coupling sector feasibility check

The Discussion in the main text anchors the hyperchaotic regime to the 2542× silicon optomechanical platform of Mayor *et al.* (2025). This section reports the same comparison parameter-by-parameter, so the order-of-magnitude gap is verifiable rather than asserted.

The physically anchored reference set (Section S2) supplies the device-level numbers: $\omega_m/2\pi = 7.436\,\mathrm{GHz}$, $g_{\mathrm{anchor}}/\omega_m \approx 1.18\times10^{-4}$, $\gamma_{\mathrm{anchor}}/\omega_m \approx 2.78\times10^{-5}$, and $\kappa_{\mathrm{anchor}}/\omega_m \approx 0.108$. The strong-coupling sector of the present manuscript uses $(g_1, g_2)/\omega_m = (0.30, 0.27)$, $\gamma_{1,2}/\omega_m = 0.02$, and $\kappa/\omega_m = 1$. The matching factor (defined as the dimensionless ratio of the present parameter over the anchored parameter) is reported per parameter in Table S18.

Table S18: Matching factor of the strong-coupling sector relative to the silicon-optomechanical anchor of Mayor *et al.* (2025). The factor is the dimensionless ratio of the dimensioned parameter used in the present work divided by the same parameter inferred from the anchored platform. Every factor lies above unity by at least one order of magnitude; the hyperchaotic regime is therefore a model-level extension, not a calibrated device prediction. The J_m/ω_m anchor entry spans the scan range of the Mathew *et al.* feasibility study.

Parameter (dimensionless ratio)	Present sector	Mayor *et al.* anchor
g_1/ω_m	0.300	1.18×10^{-4}
g_2/ω_m	0.270	1.18×10^{-4}
γ/ω_m	0.020	2.78×10^{-5}
κ/ω_m	1.000	0.108
J_m/ω_m	0.080	$[1 \times 10^{-3}, 1 \times 10^{-1}]$

Interpretation.. The matching factor ranges from 9.3× (κ) to 2542× (g_1, three orders of magnitude above the resolved-sideband single-photon coupling). The 2542× headline-sector factor is evaluated at the mapped strong-coupling point ($g_1/\omega_m = 0.30$); the feasibility-frontier scan places the robust hyperchaos onset lower, at $g_1/\omega_m \approx 0.18$, i.e. 1525× the anchor (Table S15). The hyperchaotic regime is therefore not demonstrated by the anchored silicon optomechanical platform; broader near-term platform feasibility is not established by this comparison. The main text states the regime as a model-level demonstration, and the present table makes the per-parameter basis of that statement explicit.

S16. Initial-condition robustness of the long-window sweep

The 180/180 full-map repeat (three seeds at twelve phases and five drives) reported in the main text tests the reproducibility of the positive-exponent classification across the declared finite ensemble (see Figure 3). It does not establish global attractor uniqueness or asymptotic convergence. This section documents the dependence of the classification on the initial-condition family through the classification ensemble (twelve seeds at twelve phases and three drives); the three-seed repeat of the complete map is reported in Section S6, so both cross-checks are documented in full.

Classification ensemble: twelve seeds, twelve phases, three drives.. The map was extended to twelve independent seeds at each of twelve flux phases and three drives: 432 long-window records of 1 000 000 steps (transient 100 000), 432/432 completed at the positivity threshold 1×10^{-3}. Table S20 gives the seed range of n_+ at every phase. Initial conditions are drawn i.i.d. $\mathcal{N}(0, 0.1^2 I_6)$ (six-dimensional, $\sigma = 0.1$ per component) from `default_rng(seed)`: map seeds run $20260815 + di \cdot 12 + ti$ and the robustness ensemble uses contiguous per-drive seed blocks, 432 unique seeds in all; the flank and second-coupling maps restart the same rule from base 20 271 001 and 20 270 901 respectively; every record stores its seed. The onset at $E = 1$ is phase-locked: $n_+ = 1$ at $\theta = \pm 2\pi/3$ for all 12 seeds, while every other phase gives $n_+ = 0$ (except two seeds at $\theta = -5\pi/6$). At $E = 4$ the count is seed-dependent at $\theta = 0$ ($n_+ \in \{1, 2\}$, the marginal second exponent) and robustly 1 at $|\theta| = \pi$. At $E = 8$ the count is unanimously $n_+ = 2$ for all 12 seeds at eight of the twelve phases, and $n_+ \in \{3, 4\}$ at $\theta = \pm\pi/3$ and $\theta = +\pi/2$; at $\theta = -\pi/2$ the spread is $n_+ \in \{2, 3, 4\}$. The seed-to-seed spread is at most one at eleven of twelve phases (the exception being that $\theta = -\pi/2$). The integer ranges agree at θ and $-\theta$ at all but two of the fifteen mirrored phase pairs ($E = 1$, $\theta = \mp 5\pi/6$ and $E = 8$, $\theta = \mp\pi/2$), supporting the phase-modulation statement but, as before, not establishing exact $\theta \to -\theta$ symmetry, and they do not constitute a global uniqueness proof: the ensemble is larger by a factor of four per point, not infinite. A second full twelve-phase, five-drive map at reduced coupling ($g_1/\omega_m = 0.2$, $g_2/\omega_m = 0.18$) shows the same flux modulation with a lowered ceiling: n_+ spans $\{1, 2\}$ at $E = 2.0$, $\{0, 2\}$ at $E = 4.0$ and $\{1, 3\}$ at $E = 8.0$, peaking near $\theta = \pm\pi/3$ against a flux-off baseline of 2.

Table S19: Second-coupling map ($g_1/\omega_m = 0.2$, $g_2/\omega_m = 0.18$): n_+ range over twelve phases per drive (1×10^6 steps; seeds 20 270 901–20 270 960).

E	n_+ range (12 phases)
0.2	0
1.0	0
2.0	$\{1, 2\}$
4.0	$\{0, 2\}$
8.0	$\{1, 3\}$, peaking near $\theta = \pm\pi/3$

Table S20: Seed range of the hyperchaos order n_+ over 12 seeds at each of twelve flux phases and three drives (threshold 1×10^{-3}, 1×10^6 steps; 432/432 completed). "Range" is min–max over the 12 seeds at that phase; the $-\pi$ and $+\pi$ grid ends are identified.

θ/π	$E = 1$	$E = 4$	$E = 8$
-1	0	1	2
$-5/6$	0–1	1	2
$-2/3$	1	1–2	2
$-1/2$	0	2–3	2–4
$-1/3$	0	2–3	3–4
$-1/6$	0	2	2
0	0	1–2	2
$+1/6$	0	2	2
$+1/3$	0	2–3	3–4
$+1/2$	0	2–3	3–4
$+2/3$	1	1–2	2
$+5/6$	0	1	2

S17. Local susceptibility–stretching covariation on the attractor

The main text explains the null matched-Fisher gain on the chaotic attractor with the matched-Fisher lemma: signal and noise projected onto the unstable manifold share the same stretching. This section tests that explanation directly on the trajectory, at the window scale, instead of only through the time-averaged statistic.

Method.. At six drive–phase points ($E \in \{4, 8\}$, $\theta/\pi \in \{0, 1/2, 1\}$) one trajectory of 1 000 000 steps (transient 100 000) is integrated with the same fixed-step RK4/QR backend as the production map. Alongside the state, the force-response vector $w' = J(x)\, w + e_F$, $e_F = (0, 0, 0, 1, 0, 0)$ (a constant weak force on the β_1 momentum equation), is integrated with w reset to zero at every window boundary of 2000 steps (20 time units). This is the exact window-restricted linear response $S_k = \partial_F \langle X_a \rangle$ over window k, with no finite-difference nonlinearity. Per window the integration also records the local finite-time Lyapunov exponents λ_k (QR block accumulators over the same window) and the local record spread σ_k (standard deviation of $X_a = \mathrm{Re}\,\alpha$ within the window). The window-resolved Fisher proxy is $f_k = S_k^2/\sigma_k^2$. Each point contributes 450 windows, and all six points return finite records with positive σ_k and $|S_k|$.

Shared stretching.. Signal and noise both co-vary positively with the same local stretching at every point (Table S21 and Figure S2): $\rho(\lambda_k, |S_k|)$ ranges from $+0.371$ to $+0.639$ and $\rho(\lambda_k, \sigma_k)$ from $+0.104$ to $+0.569$ (Spearman, 450 windows per point). This is the window-scale signature of the lemma's premise: the same local Lyapunov exponent that stretches the trajectory also controls both the force response and the record spread.

The two responses are not equal, however. The log-log slopes against λ_k are 7.5–13.1 for $\ln|S_k|$ but only 0.12–0.58 for $\ln\sigma_k$: within a 20-time-unit window the record spread saturates at the attractor scale while the force response keeps accumulating. The local proxy is therefore not flat—it grows with local stretching ($\rho(\lambda_k, \ln f_k) = +0.37$ to $+0.63$, coefficient of variation 6.5–13.7). This asymmetry is a new nuance of the budget picture and is reported rather than suppressed: the matched-Fisher lemma concerns the time-averaged, matched-resource statistic, whereas the local proxy here is a window-restricted diagnostic.

Flux-phase dependence of the local budget.. Two-sample Kolmogorov–Smirnov tests on the window distributions of $\ln f_k$ (450 windows per phase) give the local counterpart of the global null result. At $E = 4$, where the global matched Fisher resolves both $\theta = 0$ and $\theta = \pi$, the proxy is statistically indistinguishable between $\theta = 0$ and $\theta = \pi/2$ ($p = 0.205$) and its median is *lower* at $\theta = \pi$ than at $\theta = 0$ ($p = 0.012$): no flux phase supplies a local enhancement mechanism, and the direction matches the resolved global ratio $F_C(\pi)/F_C(0) = 0.54$ (Section S12). At $E = 8$ the $\theta = 0$ and $\theta = \pi$ distributions are statistically indistinguishable ($p = 0.766$), consistent with the global statistic being below resolution at that drive. The intermediate phase $\theta = \pi/2$ differs from $\theta = \pi$ at both drives and from $\theta = 0$ at $E = 8$ ($p \le 1 \times 10^{-5}$); the global Fisher at that phase is likewise below resolution, so the local difference cannot be converted into a global gain claim either way. These are descriptive, model-level, classical linear-response statements on the tested trajectories; no quantum, SQL, or device claim follows from them. Window-to-window lag-1 autocorrelation of $\ln f_k$ is -0.06–0.13 (effective N 347–504 vs nominal 450), leaving the reported p-values essentially unchanged; the three rejections survive Holm at $\alpha = 0.05$ over the five cross-phase comparisons.

References

[1] F. M. Mayor, S. Malik, A. G. Primo, S. Gyger, W. Jiang, T. P. M. Alegre, A. H. Safavi-Naeini, High photon-phonon pair generation rate in a two-dimensional optomechanical crystal, Nature Communications 16 (2025) 2576. doi:10.1038/s41467-025-57948-7.

Table S21: Window-resolved covariation at the six tested points. ρ is the Spearman correlation over 450 windows; the slopes are ordinary least-squares fits of the log-log response against the local finite-time Lyapunov exponent λ_k.

Point	Windows	$\rho(\lambda_k, \lvert S_k\rvert)$	$\rho(\lambda_k, \sigma_k)$	$\mathrm{d}\ln\lvert S\rvert/\mathrm{d}\lambda$	$\mathrm{d}\ln\sigma/\mathrm{d}\lambda$
$E = 4$, $\theta = 0$	450	0.440	0.104	7.5	0.12
$E = 4$, $\theta = \pi/2$	450	0.475	0.389	10.1	0.55
$E = 4$, $\theta = \pi$	450	0.371	0.116	8.4	0.18
$E = 8$, $\theta = 0$	450	0.630	0.569	10.3	0.58
$E = 8$, $\theta = \pi/2$	450	0.639	0.245	13.1	0.25
$E = 8$, $\theta = \pi$	450	0.583	0.383	10.5	0.34

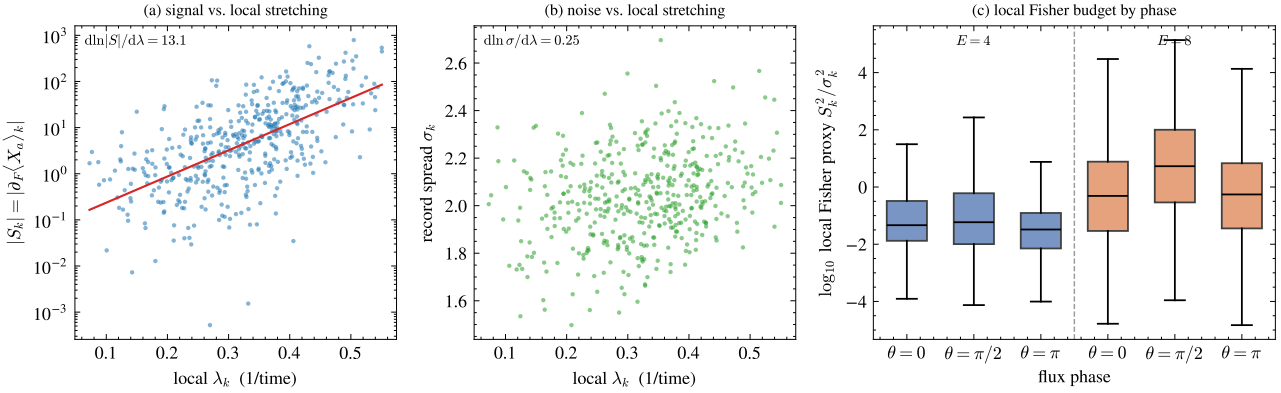


Figure S2: Window-resolved susceptibility–stretching covariation on the chaotic attractor at 450 windows per point. (a) Local force-to-observable susceptibility $|S_k|$ against the local finite-time Lyapunov exponent at $E = 8$, $\theta = \pi/2$: the signal responds strongly to local stretching. (b) Local record spread σ_k against the same exponent: the noise response is nearly flat because the within-window spread saturates at the attractor scale. (c) Distribution of the local Fisher proxy S_k^2/σ_k^2 (logarithm) at all six drive–phase points. Model-level classical linear response only.

[2] J. P. Mathew, J. del Pino, E. Verhagen, Synthetic gauge fields for phonon transport in a nano-optomechanical system, Nature Nanotechnology 15 (2020) 198–202. doi:10.1038/s41565-019-0630-8.

[3] D. Muthukumar, S. R. Mbokop Tchounda, S. T. Kingni, K. Rajagopal, S. G. Nana Engo, Controlling complex dynamics with synthetic magnetism in optomechanical systems: A route to enhanced sensor performance, Physical Review Applied 24 (1) (2025) 014053. doi:10.1103/f6sv-4fq9.

[4] G. Benettin, L. Galgani, A. Giorgilli, J.-M. Strelcyn, Lyapunov characteristic exponents for smooth dynamical systems and for hamiltonian systems; a method for computing all of them. part 2: Numerical application, Meccanica 15 (1980) 21–30. doi:10.1007/BF02128237.

[5] A. Wolf, J. B. Swift, H. L. Swinney, J. A. Vastano, Determining lyapunov exponents from a time series, Physica D 16 (1985) 285–317. doi:10.1016/0167-2789(85)90011-9.

[6] J.-P. Eckmann, D. Ruelle, Ergodic theory of chaos and strange attractors, Reviews of Modern Physics 57 (1985) 617–656. doi:10.1103/RevModPhys.57.617.

[7] A. A. Clerk, M. H. Devoret, S. M. Girvin, F. Marquardt, R. J. Schoelkopf, Introduction to quantum noise, measurement, and amplification, Reviews of Modern Physics 82 (2010) 1155–1208. doi:10.1103/RevModPhys.82.1155.

[8] S. L. Braunstein, C. M. Caves, Statistical distance and the geometry of quantum states, Physical Review Letters 72 (1994) 3439–3443. doi:10.1103/PhysRevLett.72.3439.

[9] D. F. Walls, G. J. Milburn, Quantum Optics, 2nd Edition, Springer, Berlin, 2008.

[10] C. W. Gardiner, M. J. Collett, Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation, Physical Review A 31 (1985) 3761–3774. doi:10.1103/PhysRevA.31.3761.